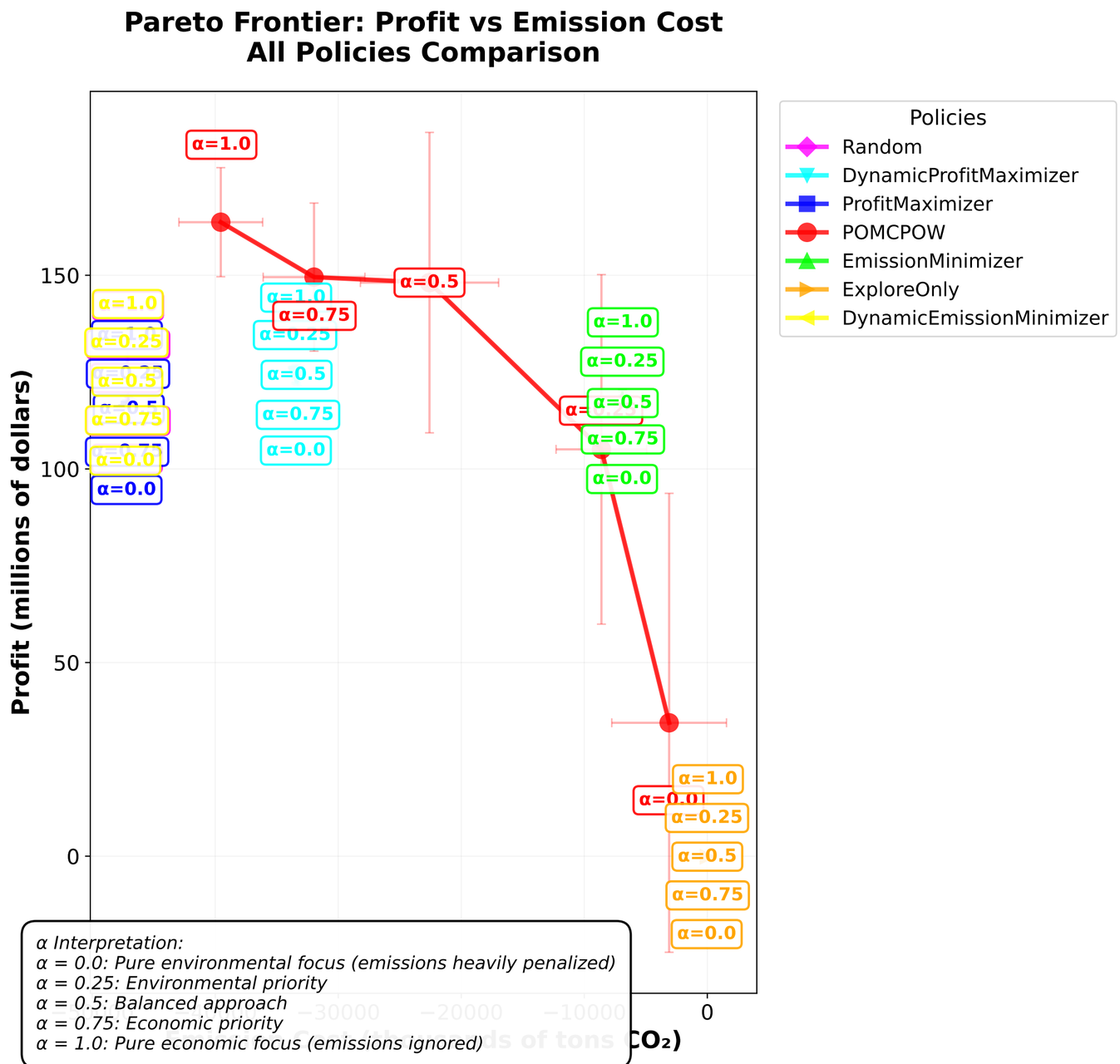
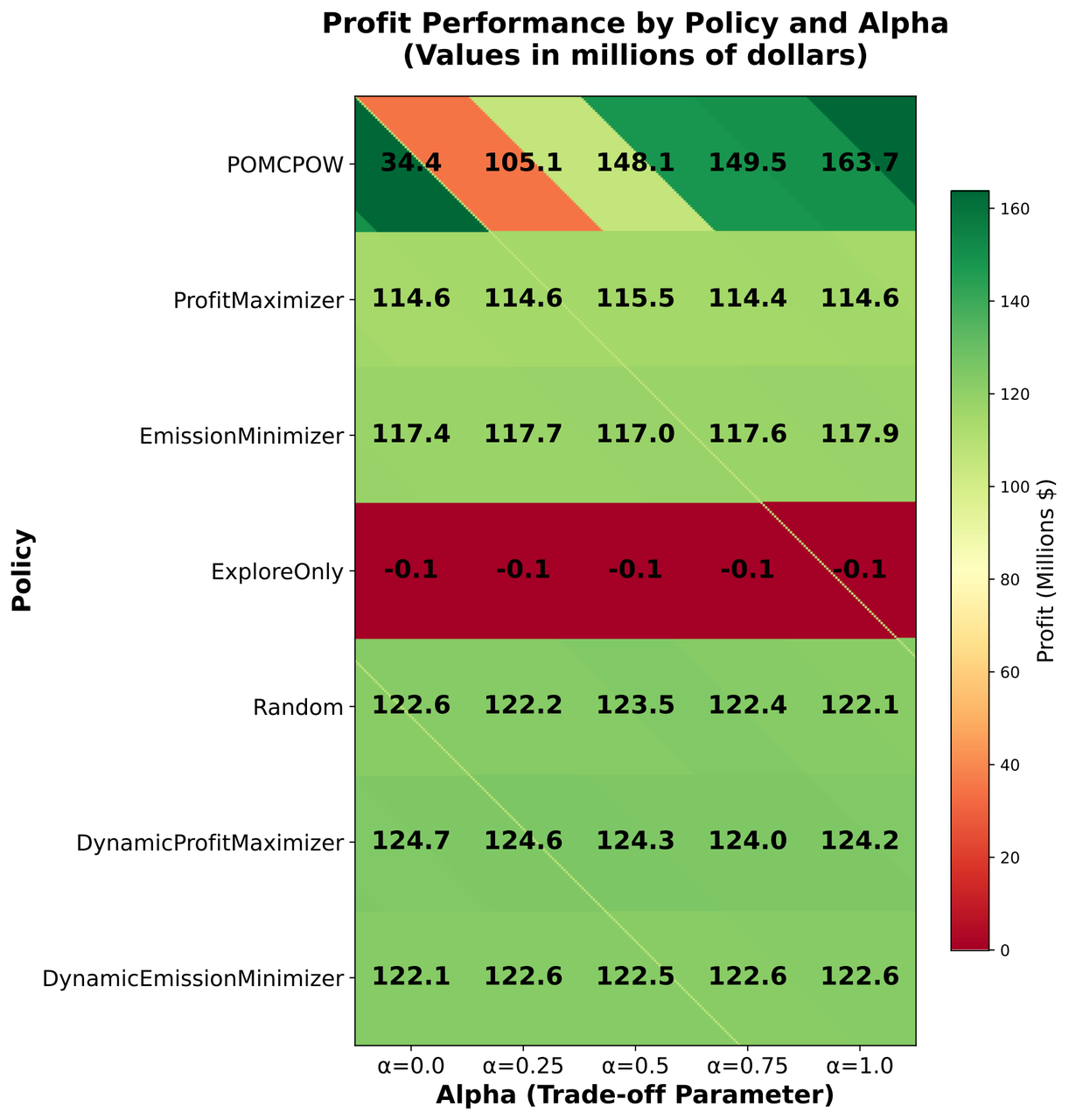

Pareto Frontier: Profit vs Emission Cost
All Policies Comparison
Profit (millions of dollars)
150
100
50
0
−30000
−20000
−10000
0
Policies
Random
DynamicProfitMaximizer
ProfitMaximizer
POMCPOW
EmissionMinimizer
ExploreOnly
DynamicEmissionMinimizer
α Interpretation:
α = 0.0: Pure environmental focus (emissions heavily penalized)
α = 0.25: Environmental priority
α = 0.5: Balanced approach
α = 0.75: Economic priority
α = 1.0: Pure economic focus (emissions ignored)


Profit Performance by Policy and Alpha
(Values in millions of dollars)
Policy
POMCPOW 34.4 105.1 148.1 149.5 163.7
ProfitMaximizer 114.6 114.6 115.5 114.4 114.6
EmissionMinimizer 117.4 117.7 117.0 117.6 117.9
ExploreOnly -0.1 -0.1 -0.1 -0.1 -0.1
Random 122.6 122.2 123.5 122.4 122.1
DynamicProfitMaximizer 124.7 124.6 124.3 124.0 124.2
DynamicEmissionMinimizer 122.1 122.6 122.5 122.6 122.6
α=0.0
α=0.25
α=0.5
α=0.75
α=1.0
Alpha (Trade-off Parameter)
Profit (Millions $)
160
140
120
100
80
60
40
20
0